

Self-Study Assignment Handout

Comparative Study of Three Constrained Optimization Methods

Methods to be used

1. Rosen's Gradient Projection Method
 2. Generalized Reduced Gradient (GRG) Method
 3. Sequential Quadratic Programming (SQP)
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Assignment Topic

Solve **one nonlinear programming problem** using the following three methods separately:

- Rosen's Gradient Projection Method
- Generalized Reduced Gradient Method
- Sequential Quadratic Programming

Then compare the results and decide **which method performs better** for the chosen problem.

Purpose of the Assignment

This assignment is designed to help you:

- understand how different constrained optimization methods work,
 - apply each method step by step,
 - compare their convergence and performance,
 - develop the ability to judge which method is more suitable for a given nonlinear problem.
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What You Have to Do

Step 1: Choose one nonlinear programming problem

Select **one** constrained nonlinear optimization problem with:

- a nonlinear objective function,
- one or more constraints,
- a clearly defined feasible starting point.

You may use:

- a textbook problem, or
 - a self-formulated real-life application problem.
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Step 2: Solve the same problem by all three methods

You must solve the **same problem** separately by:

1. Rosen's Gradient Projection Method
2. GRG Method
3. SQP Method

Do **not** use different problems for different methods.

Step 3: Use the same basis for comparison

For fair comparison, keep the following as similar as possible for all three methods:

- same problem,
 - same initial point,
 - same stopping rule,
 - same accuracy/tolerance.
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What to Include for Each Method

For **each** method, your work should contain:

1. Brief theory

Write a short note explaining:

- the basic idea of the method,
- how it chooses the search direction,
- how feasibility is maintained,
- how the next point is obtained.

2. Algorithm

Write the step-by-step algorithm in your own words.

3. Problem statement

Clearly write:

- objective function,
- constraints,
- starting point,
- any assumptions used.

4. Stepwise solution

Show the important steps of the computation, such as:

- gradients / Jacobian / Hessian (where needed),

- search direction,
- projection / reduced gradient / quadratic subproblem,
- step length,
- next iterate,
- feasibility check,
- convergence test.

5. Final result

Report:

- final optimal point,
 - final objective value,
 - whether all constraints are satisfied,
 - number of iterations used.
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Special Instructions for Each Method

A. Rosen's Gradient Projection Method

You should clearly show:

- active constraints,
- negative gradient of the objective,
- projected direction,
- movement along the feasible region,
- step-length choice.

Try to explain the geometry of the method wherever possible.

B. Generalized Reduced Gradient Method

You should clearly show:

- division of variables into basic and nonbasic variables,
 - reduced gradient calculation,
 - update of variables,
 - maintenance of feasibility,
 - convergence process.
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C. Sequential Quadratic Programming

You should clearly show:

- quadratic approximation/subproblem,
 - linearized constraints,
 - search direction,
 - Hessian or Hessian approximation,
 - update rule,
 - convergence behavior.
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Comparison Section

After completing all three solutions, compare the methods using the following points:

- final objective value,
 - feasibility of final solution,
 - number of iterations,
 - computational difficulty,
 - ease of understanding,
 - ease of implementation,
 - convergence speed,
 - overall effectiveness.
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Final Conclusion

Write a short conclusion under the heading:

Which Method Performs Better?

In this section, explain:

- which method converged faster,
- which method was easier to apply,
- which method gave the best result,
- which method you would recommend for this problem and why.

Your conclusion must be based on your actual solution, not only on theory.

Suggested Report Format

Your report should be organized in the following order:

1. Title
2. Objective
3. Problem statement
4. Solution using Rosen's Gradient Projection Method

5. Solution using GRG Method
 6. Solution using SQP Method
 7. Comparison table
 8. Final conclusion
 9. References
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Submission Guidelines

- Write neatly and systematically.
 - Show all important steps.
 - Do not skip the reasoning.
 - Tables, diagrams, and flowcharts may be added.
 - If software is used, explain the method properly; do not submit software output alone.
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Expected Learning

By doing this assignment, you should understand:

- how constrained optimization methods differ,
 - how feasibility is preserved in each method,
 - why some methods converge faster than others,
 - how to compare optimization methods logically.
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General Hint

In many smooth nonlinear constrained problems, **SQP often performs very well**, but your conclusion should come from your own calculations and observations.

Suggested Comparison Table

Method	Final Solution	Objective Value	Feasible?	Iterations	Difficulty Level	Remarks
Rosen's Gradient Projection						
GRG						
SQP						

Recommended Assignment Problem

$$\text{Minimize } f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

subject to

$$g_1(x_1, x_2) = x_1 + x_2 - 1.5 \leq 0$$

$$g_2(x_1, x_2) = x_1 - 2x_2 + 0.2 \leq 0$$

$$x_1 \geq 0, \quad x_2 \geq 0$$

with the initial feasible point

$$X^{(0)} = (0.4, 0.8)^T$$

Mode of Submission

- The assignment must be submitted in **hard copy only**.
- Students should submit a **neatly written or printed hard copy** of the full report.
- Soft copy submission will not be considered unless specifically instructed.
- All pages must be **properly arranged, stapled, and labeled** with:
 - Name of the student
 - Roll number
 - Course name/code
 - Title of the assignment

Follow the pattern of S S Rao Text Book